Operating System Interview Question

A list of top frequently asked **Operating System interview questions** and answers are given below.

1) What is an operating system?

The operating system is a software program that facilitates computer hardware to communicate and operate with the computer software. It is the most important part of a computer system without it computer is just like a box.

2) What is the main purpose of an operating system?

There are two main purposes of an operating system:

- It is designed to make sure that a computer system performs well by managing its computational activities.
- It provides an environment for the development and execution of programs.

3) What are the different operating systems?

- Batched operating systems
- Distributed operating systems
- Timesharing operating systems
- Multi-programmed operating systems
- Real-time operating systems

4) What is a socket?

A socket is used to make connection between two applications. Endpoints of the connection are called socket.

5) What is a real-time system?

Real-time system is used in the case when rigid-time requirements have been placed on the operation of a processor. It contains a well defined and fixed time constraints.

6) What is kernel?

Kernel is the core and most important part of a computer operating system which provides basic services for all parts of the OS.

7) What is monolithic kernel?

A monolithic kernel is a kernel which includes all operating system code is in single executable image.

8) What do you mean by a process?

An executing program is known as process. There are two types of processes:

- Operating System Processes
- User Processes

9) What are the different states of a process?

A list of different states of process:

- New Process
- Running Process
- Waiting Process
- Ready Process
- Terminated Process

10) What is the difference between micro kernel and macro kernel?

Micro kernel: micro kernel is the kernel which runs minimal performance affecting services for operating system. In micro kernel operating system all other operations are performed by processor.

Macro Kernel: Macro Kernel is a combination of micro and monolithic kernel.

11) What is the concept of reentrancy?

It is a very useful memory saving technique that is used for multi-programmed time sharing systems. It provides functionality that multiple users can share a single copy of program during the same period.

It has two key aspects:

- The program code cannot modify itself.
- The local data for each user process must be stored separately.

12) What is the difference between process and program?

A program while running or executing is known as a process.

13) What is the use of paging in operating system?

Paging is used to solve the external fragmentation problem in operating system. This technique ensures that the data yc need is available as quickly as possible.

14) What is the concept of demand paging?

Demand paging specifies that if an area of memory is not currently being used, it is swapped to disk to make room for a application's need.

15) What is the advantage of a multiprocessor system?

As many as processors are increased, you will get the considerable increment in throughput. It is cost effective als because they can share resources. So, the overall reliability increases.

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16) What is virtual memory?

Virtual memory is a very useful memory management technique which enables processes to execute outside of memor This technique is especially used when an executing program cannot fit in the physical memory.

17) What is thrashing?

Thrashing is a phenomenon in virtual memory scheme when the processor spends most of its time in swapping page rather than executing instructions.

18) What are the four necessary and sufficient conditions behind the deadlock?

These are the 4 conditions:

1) **Mutual Exclusion Condition**: It specifies that the resources involved are non-sharable.

2) **Hold and Wait Condition**: It specifies that there must be a process that is holding a resource already allocated to while waiting for additional resource that are currently being held by other processes.

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3) **No-Preemptive Condition**: Resources cannot be taken away while they are being used by processes.

4) **Circular Wait Condition**: It is an explanation of the second condition. It specifies that the processes in the system for circular list or a chain where each process in the chain is waiting for a resource held by next process in the chain.

19) What is a thread?

A thread is a basic unit of CPU utilization. It consists of a thread ID, program counter, register set and a stack.

20) What is FCFS?

FCFS stands for First Come, First Served. It is a type of scheduling algorithm. In this scheme, if a process requests the C first, it is allocated to the CPU first. Its implementation is managed by a FIFO queue.

21) What is SMP?

SMP stands for Symmetric MultiProcessing. It is the most common type of multiple processor system. In SMP, e processor runs an identical copy of the operating system, and these copies communicate with one another when require

22) What is RAID? What are the different RAID levels?

RAID stands for Redundant Array of Independent Disks. It is used to store the same data redundantly to improve the ove performance.

Following are the different RAID levels:

RAID 0 - Stripped Disk Array without fault tolerance

RAID 1 - Mirroring and duplexing

RAID 2 - Memory-style error-correcting codes

RAID 3 - Bit-interleaved Parity

RAID 4 - Block-interleaved Parity

RAID 5 - Block-interleaved distributed Parity

RAID 6 - P+Q Redundancy

23) What is deadlock? Explain.

Deadlock is a specific situation or condition where two processes are waiting for each other to complete so that they start. But this situation causes hang for both of them.

24) Which are the necessary conditions to achieve a deadlock?

There are 4 necessary conditions to achieve a deadlock:

- **Mutual Exclusion:** At least one resource must be held in a non-sharable mode. If any other process requests t

resource, then that process must wait for the resource to be released.

- **Hold and Wait:** A process must be simultaneously holding at least one resource and waiting for at least one resc that is currently being held by some other process.
- **No preemption:** Once a process is holding a resource (i.e. once its request has been granted), then that resc cannot be taken away from that process until the process voluntarily releases it.
- **Circular Wait:** A set of processes { P0, P1, P2, . . ., PN } must exist such that every P[i] is waiting for P[(i + 1) % 1)].

> Note: This condition implies the hold-and-wait condition, but it is easier to deal with the conditions if the four are considered separa

25) What is Banker's algorithm?

Banker's algorithm is used to avoid deadlock. It is the one of deadlock-avoidance method. It is named as Banker's algc on the banking system where bank never allocates available cash in such a manner that it can no longer satisf requirements of all of its customers.

26) What is the difference between logical address space and physical address space?

Logical address space specifies the address that is generated by CPU. On the other hand physical address space spe the address that is seen by the memory unit.

27) What is fragmentation?

Fragmentation is a phenomenon of memory wastage. It reduces the capacity and performance because space is inefficiently.

28) How many types of fragmentation occur in Operating System?

There are two types of fragmentation:

- **Internal fragmentation**: It is occurred when we deal with the systems that have fixed size allocation units.
- **External fragmentation**: It is occurred when we deal with systems that have variable-size allocation units.

29) What is spooling?

Spooling is a process in which data is temporarily gathered to be used and executed by a device, program or the systen associated with printing. When different applications send output to the printer at the same time, spooling keeps the jobs into a disk file and queues them accordingly to the printer.

30) What is the difference between internal commands and external commands?

Internal commands are the built-in part of the operating system while external commands are the separate file program

are stored in a separate folder or directory.

31) What is semaphore?

Semaphore is a protected variable or abstract data type that is used to lock the resource being used. The value semaphore indicates the status of a common resource.

There are two types of semaphore:

- Binary semaphores
- Counting semaphores

32) What is a binary Semaphore?

Binary semaphore takes only 0 and 1 as value and used to implement mutual exclusion and synchronize concurrent proce

33) What is Belady's Anomaly?

Belady's Anomaly is also called FIFO anomaly. Usually, on increasing the number of frames allocated to a process memory, the process execution is faster, because fewer page faults occur. Sometimes, the reverse happens, i.e., the ex time increases even when more frames are allocated to the process. This is Belady's Anomaly. This is true for certa reference patterns.

34) What is starvation in Operating System?

Starvation is Resource management problem. In this problem, a waiting process does not get the resources it needs fo time because the resources are being allocated to other processes.

35) What is aging in Operating System?

Aging is a technique used to avoid the starvation in resource scheduling system.

36) What are the advantages of multithreaded programming?

A list of advantages of multithreaded programming:

- Enhance the responsiveness to the users.
- Resource sharing within the process.
- Economical
- Completely utilize the multiprocessing architecture.

37) What is the difference between logical and physical address space?

Logical address specifies the address which is generated by the CPU whereas physical address specifies to the addres

is seen by the memory unit.

After fragmentation

38) What are overlays?

Overlays makes a process to be larger than the amount of memory allocated to it. It ensures that only important instruc data at any given time are kept in memory.

39) When does trashing occur?

Thrashing specifies an instance of high paging activity. This happens when it is spending more time paging instead of ex

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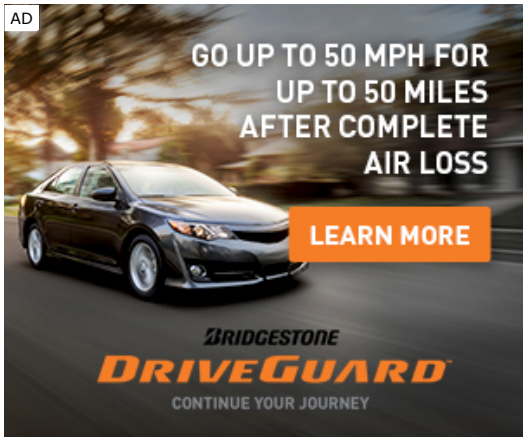

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